

Web Development Interview Questions & Answers

200 Curated Questions Across 10 Core Topics

HTML • CSS • JavaScript • React • Node.js • Databases • Web Security •
APIs & Backend • DevOps & Cloud • Advanced Web Dev

A Complete Reference Guide for Full Stack Developer Interviews

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Part 1: HTML

Q1. What is HTML?

HTML stands for HyperText Markup Language. It is the standard language used to create and structure web pages.

HTML is used to:

- Create headings
- Add paragraphs
- Insert images
- Create links
- Build forms
- Structure web content

```
<!DOCTYPE html>
<html>
<head>
  <title>My Website</title>
</head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

Q2. Difference Between HTML4 and HTML5

- HTML4: Older version | HTML5: Latest version
- HTML4: No semantic tags | HTML5: Semantic tags added
- HTML4: No direct multimedia support | HTML5: Supports audio/video
- HTML4: Less mobile friendly | HTML5: Mobile optimized

HTML5 Features:

- <video>, <audio>, <canvas>, Local Storage, Semantic Tags

Q3. What are Semantic Tags in HTML5?

Semantic tags describe the meaning of content clearly.

Common Semantic Tags:

- <header>, <nav>, <section>, <article>, <footer>

Benefits:

- Better SEO, better readability, accessibility improvement

```
<article>
  <h2>Blog Title</h2>
  <p>Content here...</p>
</article>
```

Q4. Difference Between <div> and

- <div>: Block element, takes full width, used for sections
- : Inline element, takes required width, used for small styling

```
<div>Hello</div>

<span>Hello</span>
```

Q5. What is the Purpose of DOCTYPE?

<!DOCTYPE html> tells the browser which HTML version is being used.

```
<!DOCTYPE html>
```

Benefits:

- Proper rendering and avoids browser compatibility issues

Q6. What are Meta Tags?

Meta tags provide information about the webpage.

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta name="description" content="HTML Tutorial">
```

Uses:

- SEO, responsive design, character encoding

Q7. Difference Between ID and Class

- ID: Unique, uses # in CSS
- Class: Reusable, uses . in CSS

```
<div id="header"></div>

<div class="card"></div>
<div class="card"></div>
```

Q8. What are Inline and Block Elements?

Block Elements

Start on a new line and take full width.

- Examples: <div>, <p>, <h1>

Inline Elements

Do not start on a new line; take only required space.

- Examples: , <a>,

Q9. Explain Forms in HTML

Forms collect user input.

```
<form>
  <input type="text" placeholder="Enter Name">
  <input type="email" placeholder="Enter Email">
  <button>Submit</button>
</form>
```

Common Form Elements:

- input, textarea, select, checkbox, radio button

Q10. Difference Between GET and POST

- GET: Data visible in URL, less secure, used for fetching
- POST: Data hidden, more secure, used for sending

```
<form method="GET">

<form method="POST">
```

Q11. What is localStorage and sessionStorage?

Both store data in the browser.

- localStorage: Permanent, remains after closing browser
- sessionStorage: Temporary, removed after tab closes

```
localStorage.setItem("name", "Deepak");

sessionStorage.setItem("theme", "dark");
```

Q12. What are Data Attributes?

Custom attributes used to store extra information.

```
<div data-userid="101">User</div>
```

Access in JavaScript:

```
element.dataset.userid
```

Q13. What is iframe?

iframe embeds another webpage inside a webpage.

```
<iframe src="https://example.com"></iframe>
```

Uses:

- YouTube videos, Google Maps, external websites

Q14. Difference Between Cookies and localStorage

- Cookies: Small storage, sent to server
- localStorage: Large storage, not sent automatically

```
document.cookie = "username=Deepak";
```

Q15. What are Void Elements?

Void elements do not require closing tags.

- Examples:
, <hr>, , <input>

Q16. What is the Purpose of alt Attribute?

alt provides alternative text for images.

```

```

Benefits:

- Accessibility, SEO, backup text if image fails

Q17. Explain Audio and Video Tags

HTML5 provides multimedia support.

Audio Example:

```
<audio controls>
  <source src="song.mp3">
</audio>
```

Video Example:

```
<video controls width="400">
  <source src="movie.mp4">
</video>
```

Q18. What is Accessibility in HTML?

Accessibility means making websites usable for everyone.

Best Practices:

- Semantic HTML, alt text, proper headings, keyboard support

Q19. What are ARIA Attributes?

ARIA improves accessibility for screen readers.

```
<button aria-label="Search">
```

Common ARIA Attributes:

- aria-label, aria-hidden, aria-expanded

Q20. Difference Between and

- : Semantic importance, used for important text
- : Only bold styling, used for design

```
<strong>Important</strong>
```

```
<b>Bold Text</b>
```

Part 2: CSS

Q21. What is CSS?

CSS stands for Cascading Style Sheets. It is used to style and design HTML elements.

CSS helps to:

- Add colors, set fonts, create layouts, add animations, make websites responsive

```
h1 {  
  color: blue;  
  font-size: 40px;  
}
```

Q22. Difference Between Inline, Internal, and External CSS

- Inline CSS: Written inside HTML element
- Internal CSS: Written inside <style> tag
- External CSS: Written in a separate .css file

```
<h1 style="color:red;">Hello</h1>  
  
<style>  
h1 {  
  color: blue;  
}  
</style>  
  
<link rel="stylesheet" href="style.css">
```

Q23. What is Specificity in CSS?

Specificity determines which CSS rule is applied when multiple rules target the same element.

Priority Order:

- 1. Inline CSS
- 2. ID Selector
- 3. Class Selector
- 4. Element Selector

```
#title {  
  color: red;  
}  
  
.heading {  
  color: blue;  
}
```

ID selector has higher priority.

Q24. Explain CSS Box Model

Every HTML element is treated as a box.

The box model contains:

- Content, Padding, Border, Margin (Margin > Border > Padding > Content)

```
div {  
  padding: 20px;  
  border: 2px solid black;  
  margin: 10px;  
}
```

Q25. Difference Between Margin and Padding

- Margin: Space outside border, creates gap between elements
- Padding: Space inside border, creates inner spacing

```
div {  
  margin: 20px;  
  padding: 20px;  
}
```

Q26. What is Flexbox?

Flexbox is a one-dimensional layout system used for alignment and spacing.

Benefits:

- Easy alignment, responsive layouts, flexible spacing

```
.container {  
  display: flex;  
  justify-content: center;  
  align-items: center;  
}
```

Q27. What is CSS Grid?

CSS Grid is a two-dimensional layout system that handles rows and columns.

```
.container {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
}
```

Q28. Difference Between Relative, Absolute, Fixed, and Sticky Positioning

- relative: Positioned relative to itself
- absolute: Positioned relative to parent
- fixed: Fixed on screen
- sticky: Sticks during scrolling

```
div {  
  position: absolute;  
  top: 20px;  
}
```

Q29. What is z-index?

z-index controls the stack order of elements. Higher z-index appears on top.

```
.box {  
  z-index: 10;  
}
```

Q30. Difference Between em, rem, %, px, vh, and vw

- px: Fixed pixels
- %: Relative percentage
- em: Relative to parent
- rem: Relative to root
- vh: Viewport height
- vw: Viewport width

```
h1 {  
  font-size: 2rem;  
}
```

Q31. What are Pseudo-Classes?

Pseudo-classes define special states of elements.

```
a:hover {  
  color: red;  
}
```

Common Pseudo-Classes:

- :hover, :focus, :first-child, :last-child

Q32. What are Pseudo-Elements?

Pseudo-elements style specific parts of elements.

```
p::first-letter {  
  font-size: 40px;  
}
```

Common Pseudo-Elements:

- ::before, ::after, ::first-letter

Q33. Difference Between visibility:hidden and display:none

- visibility:hidden: Element hidden but space remains
- display:none: Element removed completely

```
.box {  
  display: none;  
}
```

Q34. What is Media Query?

Media queries make websites responsive.

```
@media (max-width: 768px) {  
  body {  
    background: lightblue;  
  }  
}
```

Q35. Explain Responsive Design

Responsive design ensures websites work on mobile, tablet, and desktop.

Techniques:

- Media queries, flexible layouts, responsive images

Q36. What is Mobile-First Design?

Mobile-first design means designing for smaller screens first and then scaling upward.

Benefits:

- Better performance, better UX on mobile devices

Q37. What are CSS Transitions?

Transitions create smooth changes between property values.

```
button {  
  transition: background 0.3s ease;  
}
```

Q38. Difference Between Transition and Animation

- Transition: Triggered by event, simple effects
- Animation: Can run automatically, complex effects

```
@keyframes move {  
  from { left:0; }  
  to { left:100px; }  
}
```

Q39. What is transform Property?

transform changes element shape or position.

```
div {  
  transform: rotate(45deg);  
}
```

Common Functions:

- rotate(), scale(), translate()

Q40. What is overflow Property?

Controls content overflow behavior.

Values:

- visible, hidden, scroll, auto

```
div {  
  overflow: scroll;  
}
```

Q41. Explain CSS Inheritance

Some CSS properties automatically pass from parent to child.

```
body {  
  color: blue;  
}
```

Child elements inherit text color.

Q42. What is !important?

!important gives highest priority to a CSS rule.

```
p {  
  color: red !important;  
}
```

Q43. What are CSS Preprocessors?

Preprocessors extend CSS functionality.

Popular Preprocessors:

- SASS, SCSS, LESS

Features:

- Variables, nesting, functions

Q44. Difference Between SCSS and SASS

- SCSS: Uses braces, CSS-like syntax
- SASS: No braces, indentation syntax

```
$color: blue;  
  
h1 {  
  color: $color;  
}
```

Q45. What is Bootstrap?

Bootstrap is a popular CSS framework used for responsive web development.

Features:

- Responsive grid system, prebuilt components, faster development

```
<button class="btn btn-primary">  
  Click Me  
</button>
```

Part 3: JavaScript

Q46. What is JavaScript?

JavaScript is a programming language used to make web pages interactive.

It is used for:

- Form validation, dynamic content, API calls, animations, interactive UI

```
console.log("Hello World");
```

Q47. Difference Between var, let, and const

- var: Function scoped, can redeclare, can reassign
- let: Block scoped, cannot redeclare, can reassign
- const: Block scoped, cannot redeclare, cannot reassign

```
var a = 10;  
let b = 20;  
const c = 30;
```

Q48. What are Data Types in JavaScript?

Primitive Data Types:

- String, Number, Boolean, Null, Undefined, Symbol, BigInt

Non-Primitive:

- Object, Array, Function

```
let name = "Sanyam";  
let age = 26;  
let isActive = true;
```

Q49. Difference Between null and undefined

- null: Intentional empty value
- undefined: Variable declared but not assigned

```
let a = null;  
let b;  
  
console.log(b);
```

Q50. What is Hoisting?

Hoisting means JavaScript moves declarations to the top before execution.

```
console.log(a);

var a = 10;
// Output: undefined
```

Q51. Explain Scope in JavaScript

Scope determines where variables are accessible.

Types:

- Global Scope, Function Scope, Block Scope

```
{
  let age = 26;
}
// age cannot be accessed outside block.
```

Q52. What is Closure?

A closure allows a function to access variables from its outer function even after execution.

```
function outer() {
  let count = 0;

  return function inner() {
    count++;
    console.log(count);
  };
}

const fn = outer();
fn();
```

Q53. What is Callback Function?

A callback is a function passed into another function.

```
function greet(name, callback) {
  console.log(name);
  callback();
}

greet("Deepak", function() {
  console.log("Welcome");
});
```

Q54. What is Arrow Function?

Arrow functions provide shorter syntax.

```
const add = (a, b) => a + b;
```

Benefits:

- Cleaner syntax, lexical this

Q55. Difference Between == and ===

- ==: Checks value only
- ===: Checks value and type

```
5 == "5"    // true
5 === "5"   // false
```

Q56. What is Event Bubbling?

Event bubbling means events move from child to parent.

```
button.addEventListener("click", () => {
  console.log("Button clicked");
});
```

Q57. What is Event Capturing?

Event capturing is the opposite of bubbling. Event moves from Parent to Child.

```
element.addEventListener("click", fn, true);
```

Q58. What is Event Delegation?

Using a parent element to handle child events.

Benefits:

- Better performance, dynamic elements support

```
parent.addEventListener("click", (e) => {
  console.log(e.target);
});
```

Q59. What is DOM?

DOM stands for Document Object Model. It represents HTML as objects.

```
document.getElementById("title");
```

Q60. Difference Between Synchronous and Asynchronous Programming

- Synchronous: Executes line by line, blocking
- Asynchronous: Executes independently, non-blocking

```
setTimeout(() => {  
  console.log("Hello");  
}, 2000);
```

Q61. What is Promise in JavaScript?

Promise handles asynchronous operations.

States:

- Pending, Resolved, Rejected

```
const promise = new Promise((resolve, reject) => {  
  resolve("Success");  
});
```

Q62. What are async and await?

Used to simplify asynchronous code.

```
async function getData() {  
  const response = await fetch(url);  
}
```

Q63. What is setTimeout?

Runs code after a specific delay.

```
setTimeout(() => {  
  console.log("Hello");  
}, 3000);
```

Q64. Difference Between map(), filter(), and reduce()

- map(): Transform array
- filter(): Filter array
- reduce(): Reduce array to single value

```
const nums = [1,2,3];  
  
nums.map(n => n * 2);
```

Q65. What is Destructuring?

Extract values from arrays or objects.

```
const person = {  
  name: "Deepak"  
};  
  
const { name } = person;
```

Q66. What is Spread Operator?

Spread operator expands elements.

```
const arr1 = [1,2];  
const arr2 = [...arr1, 3];
```

Q67. What is Rest Operator?

Collects multiple values into an array.

```
function sum(...nums) {  
  console.log(nums);  
}
```

Q68. What is Template Literal?

Template literals allow embedded expressions.

```
let name = "Narayan";  
  
console.log(`Hello ${name}`);
```

Q69. What is Optional Chaining?

Safely accesses nested properties.

```
user?.address?.city
```

Q70. Explain this Keyword

this refers to the current object.

```
const user = {  
  name: "Radhe",  
  greet() {  
    console.log(this.name);  
  }  
};
```

Q71. Difference Between Function Declaration and Function Expression

- Declaration: Hoisted
- Expression: Not fully hoisted

```
function test() {}  
  
const demo = function() {};
```

Q72. What is Prototype?

Prototype allows inheritance in JavaScript.

```
function Person() {}  
  
Person.prototype.age = 25;
```

Q73. What is Prototypal Inheritance?

Objects inherit properties from other objects.

```
child.__proto__ = parent;
```

Q74. What is JSON?

JSON stands for JavaScript Object Notation. Used for data exchange.

```
{  
  "name": "Sanyam",  
  "age": 26  
}
```

Q75. Difference Between localStorage and sessionStorage

- localStorage: Permanent, shared across tabs
- sessionStorage: Temporary, limited to tab

Q76. What is NaN?

NaN means Not a Number.

```
console.log("abc" / 2);
```

Q77. What are Truthy and Falsy Values?

Falsy Values:

- false, 0, "", null, undefined, NaN

Everything else is truthy.

Q78. What is Debounce?

Debounce limits repeated function calls.

Used in:

- Search bars, resize events

```
function debounce(fn, delay) {}
```

Q79. What is Throttle?

Throttle limits execution rate.

```
function throttle(fn, limit) {}
```

Q80. What is Currying?

Currying converts a function with multiple arguments into nested functions.

```
function add(a) {  
  return function(b) {  
    return a + b;  
  };  
}
```

Part 4: ReactJS

Q81. What is React?

React is a JavaScript library used to build user interfaces. It was developed by Meta.

Features:

- Component-based architecture, Virtual DOM, reusable UI components, fast rendering

```
function App() {  
  return <h1>Hello React</h1>;  
}
```

Q82. What are Components in React?

Components are reusable building blocks of UI.

Types:

- Functional Components, Class Components

```
function Welcome() {  
  return <h1>Welcome</h1>;  
}
```

Q83. Difference Between Functional and Class Components

- Functional Component: Simpler syntax, uses hooks, preferred nowadays
- Class Component: More complex, uses lifecycle methods, older approach

```
function App() {  
  return <h1>Hello</h1>;  
}
```

Q84. What is JSX?

JSX stands for JavaScript XML. It allows writing HTML inside JavaScript.

```
const element = <h1>Hello</h1>;
```

Benefits:

- Cleaner syntax, easier UI development

Q85. What are Props?

Props are used to pass data between components.

```
function User(props) {  
  return <h1>{props.name}</h1>;  
}  
  
// Usage:  
<User name="Deepak" />
```

Q86. What is State?

State stores dynamic data inside components.

```
const [count, setCount] = useState(0);
```

Uses:

- Form handling, counters, dynamic UI updates

Q87. Difference Between State and Props

- State: Managed inside component, mutable
- Props: Passed from parent, immutable

Q88. What is useState Hook?

useState manages state in functional components.

```
import { useState } from "react";  
  
function Counter() {  
  const [count, setCount] = useState(0);  
  
  return (  
    <button onClick={() => setCount(count + 1)}>  
      {count}  
    </button>  
  );  
}
```

Q89. What is useEffect Hook?

useEffect handles side effects.

```
useEffect(() => {  
  console.log("Component Loaded");  
}, []);
```

Uses:

- API calls, timers, event listeners

Q90. What is Virtual DOM?

Virtual DOM is a lightweight copy of the real DOM. React updates only changed parts instead of the entire page.

Benefits:

- Faster updates, better performance

Q91. What is Reconciliation?

Reconciliation is React's process of comparing the old Virtual DOM and new Virtual DOM, then updating only changed elements.

Q92. What are Keys in React?

Keys uniquely identify list items.

```
items.map(item => (  
  <li key={item.id}>{item.name}</li>  
));
```

Benefits:

- Better rendering, efficient updates

Q93. What is Prop Drilling?

Passing props through multiple nested components unnecessarily.

Problem:

App -> Parent -> Child -> GrandChild

Solution:

- Context API, Redux

Q94. What is Context API?

Context API shares data globally without prop drilling.

```
const UserContext = createContext();
```

Uses:

- Theme management, authentication, global settings

Q95. What is Redux?

Redux is a state management library used in React applications.

Concepts:

- Store, Actions, Reducers

Benefits:

- Centralized state, predictable state updates

Q96. Difference Between Redux and Context API

- Redux: Advanced state management, better for large apps
- Context API: Simple global state, better for smaller apps

Q97. What are React Hooks?

Hooks allow functional components to use state and lifecycle features.

Common Hooks:

- useState, useEffect, useRef, useMemo, useCallback

Q98. What is useRef?

useRef stores mutable values without re-rendering.

```
const inputRef = useRef();
```

Uses:

- Access DOM elements, store previous values

Q99. What is useMemo?

useMemo optimizes expensive calculations.

```
const result = useMemo(() => {  
  return calculate(data);  
}, [data]);
```

Q100. What is useCallback?

useCallback memoizes functions.

```
const memoFn = useCallback(() => {  
  console.log("Hello");  
}, []);
```

Q101. What is Lazy Loading in React?

Lazy loading loads components only when needed.

```
const Home = React.lazy(() => import("./Home"));
```

Benefits:

- Faster loading, better performance

Q102. What is React Router?

React Router handles navigation in React applications.

```
<Route path="/about" element={<About />} />
```

Q103. What are Controlled Components?

Form elements controlled by React state.

```
<input  
  value={name}  
  onChange={(e) => setName(e.target.value)}  
/>
```

Q104. What are Uncontrolled Components?

Form elements managed by the DOM itself.

```
<input ref={inputRef} />
```

Q105. What is Lifting State Up?

Moving shared state to a common parent component.

Benefits:

- Better state sharing, improved synchronization

Q106. What is Higher Order Component (HOC)?

HOC is a function that takes a component and returns an enhanced component.

```
const Enhanced = withAuth(Component);
```

Q107. What are Custom Hooks?

Custom hooks reuse logic across components.

```
function useFetch() {  
  // logic  
}
```

Q108. What is Strict Mode?

React Strict Mode helps identify potential issues.

```
<React.StrictMode>  
  <App />  
</React.StrictMode>
```

Q109. What is Server-Side Rendering (SSR)?

SSR renders React components on the server before sending to the browser.

Benefits:

- Better SEO, faster initial load

Q110. Difference Between CSR and SSR

- CSR: Rendered in browser, slower initial load, SEO less optimized
- SSR: Rendered on server, faster initial load, better SEO

Part 5: Node.js

Q111. What is Node.js?

Node.js is a JavaScript runtime built on Chrome's V8 engine. It allows JavaScript to run outside the browser.

Features:

- Fast execution, event-driven, non-blocking I/O, scalable applications

```
console.log("Hello Node.js");
```

Q112. Why Use Node.js?

Advantages:

- Fast performance, single language for frontend and backend, handles multiple requests efficiently, huge npm ecosystem

Best Use Cases:

- APIs, real-time apps, chat applications, streaming services

Q113. What is npm?

npm stands for Node Package Manager. Used to install libraries/packages.

```
npm install express
```

Uses:

- Install packages, manage dependencies, run scripts

Q114. Difference Between CommonJS and ES Modules

- CommonJS: Uses `require()`, uses `module.exports`
- ES Modules: Uses `import`, uses `export`

```
const fs = require("fs");    // CommonJS
import fs from "fs";        // ES Modules
```

Q115. What is Express.js?

Express.js is a minimal backend framework for Node.js.

Features:

- Routing, middleware support, API development

```
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello");
});
```

Q116. What is Middleware?

Middleware functions execute between Request and Response.

Uses:

- Authentication, logging, validation

```
app.use((req, res, next) => {
  console.log("Middleware");
  next();
});
```

Q117. What is REST API?

REST API follows REST architecture principles.

Common Methods:

- GET, POST, PUT, DELETE

```
app.get("/users", (req, res) => {
  res.json(users);
});
```

Q118. Difference Between PUT and PATCH

- PUT: Updates entire resource
- PATCH: Updates partial resource

```
PUT /user/1
PATCH /user/1
```

Q119. What is JWT?

JWT stands for JSON Web Token. Used for authentication.

Structure:

Header.Payload.Signature

Benefits:

- Secure authentication, stateless sessions

Q120. What is Authentication vs Authorization?

- Authentication: Verifies identity
- Authorization: Verifies permissions

Example: Login -> Authentication; Admin access -> Authorization

Q121. What is CORS?

CORS stands for Cross-Origin Resource Sharing. It controls resource sharing between different domains.

```
app.use(cors());
```

Q122. What is dotenv?

dotenv loads environment variables from a .env file.

```
require("dotenv").config();

// .env file
PORT=5000
```

Q123. What is Event Loop?

Event loop handles asynchronous operations in Node.js.

Process:

1. Executes synchronous code
2. Handles callbacks
3. Processes async tasks

Benefits:

- Non-blocking execution, efficient concurrency

Q124. What is Non-Blocking I/O?

Node.js can process multiple requests without waiting.

Benefits:

- Faster performance, better scalability

Q125. What is package.json?

package.json stores project metadata and dependencies.

```
{
  "name": "myapp",
  "version": "1.0.0"
}
```

Q126. What is nodemon?

nodemon automatically restarts the server after code changes.

```
npm install -g nodemon
```

Q127. What are Streams in Node.js?

Streams process data piece by piece instead of loading all at once.

Types:

- Readable, Writable, Duplex, Transform

Benefits:

- Memory efficient, faster processing

Q128. What is Buffering?

Buffer temporarily stores binary data in memory.

```
const buffer = Buffer.from("Hello");
```

Q129. What is Async Middleware?

Middleware using async/await.

```
app.get("/", async (req, res) => {
  const data = await fetchData();
  res.json(data);
});
```

Q130. What is Rate Limiting?

Rate limiting restricts the number of requests from users.

Benefits:

- Prevents abuse, protects APIs, improves security

```
const rateLimit = require("express-rate-limit");
```

Part 6: Database

Q131. What is SQL?

SQL stands for Structured Query Language. It is used to manage and manipulate relational databases.

Uses:

- Store data, retrieve data, update records, delete records

```
SELECT * FROM users;
```

Q132. Difference Between SQL and NoSQL

- SQL: Relational database, uses tables, structured schema (e.g. MySQL)
- NoSQL: Non-relational database, uses collections/documents, flexible schema (e.g. MongoDB)

Q133. What is Primary Key?

Primary key uniquely identifies each record in a table.

Features:

- Unique, cannot be NULL

```
CREATE TABLE users (  
    id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

Q134. What is Foreign Key?

Foreign key creates a relationship between two tables.

```
CREATE TABLE orders (  
    order_id INT,  
    user_id INT,  
    FOREIGN KEY (user_id) REFERENCES users(id)  
);
```

Q135. What is Normalization?

Normalization organizes a database to reduce redundancy.

Normal Forms:

- 1NF, 2NF, 3NF

Benefits:

- Reduced duplication, better consistency, improved integrity

Q136. What are Joins in SQL?

Joins combine data from multiple tables.

Types:

- INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN

```
SELECT users.name, orders.amount
FROM users
INNER JOIN orders
ON users.id = orders.user_id;
```

Q137. Difference Between INNER JOIN and LEFT JOIN

- INNER JOIN: Returns matching rows only
- LEFT JOIN: Returns all left table rows

```
SELECT * FROM users
LEFT JOIN orders
ON users.id = orders.user_id;
```

Q138. What is Indexing?

Index improves database query performance.

Benefits:

- Faster searches, faster filtering

```
CREATE INDEX idx_name
ON users(name);
```

Q139. What is Aggregate Function?

Aggregate functions perform calculations on multiple rows.

Common Functions:

- COUNT(), SUM(), AVG(), MIN(), MAX()

```
SELECT COUNT(*) FROM users;
```

Q140. Difference Between DELETE, DROP, and TRUNCATE

- DELETE: Removes rows, can use WHERE
- DROP: Removes table, deletes structure
- TRUNCATE: Removes all rows, faster than DELETE

```
DELETE FROM users WHERE id = 1;
```

Q141. What is MongoDB?

MongoDB is a NoSQL database that stores data in JSON-like documents.

Features:

- Flexible schema, high scalability, fast performance

```
{
  "name": "Deepak",
  "age": 25
}
```

Q142. Difference Between MongoDB and MySQL

- MongoDB: NoSQL, flexible schema, document-based
- MySQL: SQL, fixed schema, table-based

Q143. What is Schema?

Schema defines the structure of a database.

```
CREATE TABLE users (
  id INT,
  name VARCHAR(50)
);
```

Q144. What is ORM?

ORM stands for Object Relational Mapping. ORM allows interaction with a database using programming language objects.

Benefits:

- Easier queries, cleaner code, faster development

Q145. What is Sequelize?

Sequelize is an ORM for Node.js.

```
User.findAll();
```

Benefits:

- Easy database interaction, supports SQL databases

Q146. What is Mongoose?

Mongoose is an ODM library for MongoDB.

```
const User = mongoose.model("User", userSchema);
```

Q147. What are ACID Properties?

ACID ensures reliable database transactions.

Properties:

- Atomicity, Consistency, Isolation, Durability

Benefits:

- Data reliability, transaction safety

Q148. What is Transaction?

Transaction is a group of database operations executed together.

```
BEGIN;  
UPDATE accounts  
SET balance = balance - 500  
WHERE id = 1;  
COMMIT;
```

Q149. What is Database Sharding?

Sharding splits a database into smaller parts.

Benefits:

- Better scalability, faster performance

Q150. What is Replication?

Replication copies database data across multiple servers.

Benefits:

- High availability, backup support, fault tolerance

Part 7: Web Security

Q151. What is HTTPS?

HTTPS stands for HyperText Transfer Protocol Secure. It is the secure version of HTTP that encrypts data exchanged between browser and server.

Benefits:

- Secure communication, protects sensitive data, prevents eavesdropping

Q152. Difference Between HTTP and HTTPS

- Security: HTTP not secure | HTTPS secure
- Data: HTTP sent as plain text | HTTPS data encrypted
- Port: HTTP uses Port 80 | HTTPS uses Port 443
- SSL/TLS: HTTP none | HTTPS requires SSL/TLS

Q153. What is SSL/TLS?

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are protocols that encrypt data transmission.

Benefits:

- Data encryption, authentication, data integrity

Q154. What is XSS Attack?

XSS stands for Cross-Site Scripting. It occurs when attackers inject malicious JavaScript into webpages.

```
<script>
alert("Hacked");
</script>
```

Prevention:

- Validate input, escape output, use Content Security Policy (CSP)

Q155. What is CSRF Attack?

CSRF stands for Cross-Site Request Forgery. It tricks authenticated users into performing unwanted actions.

Prevention:

- CSRF Tokens, SameSite Cookies, authentication checks

Q156. What is SQL Injection?

SQL Injection occurs when malicious SQL code is inserted into queries.

```
SELECT * FROM users
WHERE username = 'admin'
AND password = '123';
```

Prevention:

- Prepared Statements, parameterized queries, input validation

Q157. How to Secure APIs?

Best Practices:

- Use HTTPS, authentication & authorization, rate limiting, input validation, API keys, JWT tokens

Q158. What is Hashing?

Hashing converts data into a fixed-length value.

Uses:

- Password storage, data verification

Q159. Difference Between Encryption and Hashing

- Encryption: Reversible, uses key, protects data (e.g. credit card data)
- Hashing: Irreversible, no key required, verifies integrity (e.g. passwords)

Q160. What is bcrypt?

bcrypt is a password hashing algorithm.

Features:

- Salt generation, secure password storage, resistant to brute-force attacks

```
const hash = await bcrypt.hash(password, 10);
```

Q161. What is OAuth?

OAuth is an authorization framework that allows third-party applications to access resources without sharing passwords.

Examples:

- Login with Google, Login with GitHub, Login with Facebook

Q162. What is JWT Token Security?

JWT (JSON Web Token) securely transmits user information.

Structure:

Header.Payload.Signature

Security Tips:

- Short expiration time, use HTTPS, store securely

Q163. What is Content Security Policy (CSP)?

CSP is a browser security feature that helps prevent XSS attacks.

```
Content-Security-Policy: default-src 'self';
```

Q164. What is Brute Force Attack?

A brute force attack tries many password combinations until the correct one is found.

Prevention:

- Strong passwords, account lockout, rate limiting, multi-factor authentication

Q165. What is Two-Factor Authentication (2FA)?

2FA requires two forms of verification before granting access.

Example:

- 1. Password, 2. OTP sent to phone

Part 8: APIs & Backend Concepts

Q166. What is an API?

API stands for Application Programming Interface. An API allows different software applications to communicate with each other.

Benefits:

- Data sharing, system integration, automation

Q167. Difference Between REST and GraphQL

- REST: Multiple endpoints, fixed response structure, over-fetching possible, simpler
- GraphQL: Single endpoint, client chooses data, fetch only needed data, more flexible

```
REST:      GET /users/1
GraphQL:   { user(id:1){ name email } }
```

Q168. What is a RESTful API?

RESTful API follows REST architectural principles.

Key Principles:

- Stateless, client-server architecture, uniform interface, resource-based URLs

```
GET /users
POST /users
PUT /users/1
DELETE /users/1
```

Q169. What are HTTP Status Codes?

HTTP status codes indicate server response status.

Categories:

- 1xx Informational, 2xx Success, 3xx Redirection, 4xx Client Error, 5xx Server Error

Q170. Explain 200, 201, 400, 401, 403, 404, and 500

- 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 500 Internal Server Error

```
res.status(404).json({
  message: "User Not Found"
});
```

Q171. What is API Testing?

API Testing verifies functionality, performance, and security of APIs.

Checks:

- Response data, status codes, authentication, performance

Tools:

- Postman, Insomnia, Swagger

Q172. What is Postman?

Postman is a tool used to test APIs.

Features:

- Send API requests, test endpoints, automate testing, manage collections

Q173. What is WebSocket?

WebSocket enables real-time two-way communication between client and server.

Uses:

- Chat applications, online gaming, live notifications, stock market updates

Q174. Difference Between Polling and WebSocket

- Polling: Client repeatedly asks server, higher network usage, slower updates
- WebSocket: Persistent connection, efficient communication, real-time updates

Q175. What is Microservice Architecture?

Microservices divide an application into small independent services.

Example:

- User Service, Payment Service, Order Service, Notification Service

Benefits:

- Scalability, independent deployment, easier maintenance

Q176. What is Monolithic Architecture?

Monolithic architecture combines all components (frontend, backend, database) into one application.

Advantages:

- Simpler development initially

Disadvantages:

- Harder to scale

Q177. What is Caching?

Caching stores frequently accessed data temporarily for faster retrieval.

Benefits:

- Faster performance, reduced database load, improved user experience

```
cache.set("users", data);
```

Q178. What is Redis?

Redis is an in-memory database commonly used for caching.

Features:

- Extremely fast, key-value storage, session management

```
redis.set("user", userData);
```

Q179. What is CDN?

CDN stands for Content Delivery Network. A CDN stores copies of content across multiple servers worldwide.

Benefits:

- Faster content delivery, reduced latency, better availability

Examples:

- Cloudflare, Akamai

Q180. What is Load Balancing?

Load balancing distributes traffic across multiple servers.

Benefits:

- High availability, better performance, fault tolerance

Part 9: Deployment, DevOps & Cloud

Q181. What is Git?

Git is a distributed version control system used to track code changes.

Benefits:

- Track code history, collaborate with teams, manage versions

```
git init
```

Q182. Difference Between Git and GitHub

- Git: Version control system, works locally, tracks code changes
- GitHub: Cloud hosting platform, works online, stores repositories

Q183. What is Version Control?

Version control is a system that records changes to files over time.

Benefits:

- Roll back changes, team collaboration, change tracking

```
git log
```

Q184. What is Branching in Git?

A branch is an independent line of development.

Common Branches:

- main, develop, feature branches

```
git branch feature-login
```

Benefits:

- Safe development, parallel work

Q185. What is a Merge Conflict?

A merge conflict occurs when Git cannot automatically merge changes (e.g. two developers modify the same line of code).

Resolution Steps:

- 1. Identify conflict, 2. Edit conflicting code, 3. Commit changes

```
git add .  
git commit
```

Q186. What is CI/CD?

CI/CD stands for Continuous Integration (CI) and Continuous Deployment/Delivery (CD).

Continuous Integration:

Automatically tests code after every commit.

Continuous Deployment:

Automatically deploys code after successful testing.

Benefits:

- Faster releases, reduced bugs, automation

Q187. What is Docker?

Docker is a platform used to package applications into containers.

Benefits:

- Consistent environments, easy deployment, lightweight containers

```
docker build .
```

Q188. What is Kubernetes?

Kubernetes is a container orchestration platform.

It manages:

- Containers, scaling, load balancing, deployments

Benefits:

- Auto-scaling, high availability, container management

```
kubectl get pods
```

Q189. What is Cloud Hosting?

Cloud hosting means hosting applications on cloud servers rather than physical servers.

Benefits:

- Scalability, high availability, pay-as-you-go pricing

Popular Cloud Providers:

- AWS, Microsoft Azure, Google Cloud Platform

Q190. Difference Between AWS, Azure, and GCP

- AWS: Market leader, largest service portfolio, most popular cloud platform
- Azure: Strong Microsoft integration, enterprise-focused, popular in enterprises
- GCP: Strong AI & Analytics, developer-friendly, strong data services

AWS Popular Services:

- EC2, S3, RDS, Lambda

Azure Popular Services:

- Virtual Machines, Azure SQL, Azure Functions

GCP Popular Services:

- Compute Engine, BigQuery, Cloud Functions

Part 10: Advanced Web Development

Q191. What is a Progressive Web App (PWA)?

A Progressive Web App (PWA) is a web application that provides an app-like experience in a browser.

Features:

- Works offline, fast loading, installable on devices, push notifications, responsive design

Examples:

- Twitter Lite, Pinterest, Starbucks PWA

Benefits:

- No app store required, better performance, improved user engagement

Q192. What is a Service Worker?

A Service Worker is a JavaScript file that runs in the background independently of a webpage.

Responsibilities:

- Caching resources, offline support, push notifications, background sync

```
navigator.serviceWorker.register("/service-worker.js")
```

Benefits:

- Offline access, faster page loading, better user experience

Q193. What is Lazy Loading?

Lazy loading means loading resources only when they are needed (images, components, videos).

React Example:

```
const Home = React.lazy(() => import("./Home"))
```

Benefits:

- Faster initial page load, better performance, reduced bandwidth usage

Q194. What is Code Splitting?

Code splitting divides a large JavaScript bundle into smaller chunks.

Example:

Instead of loading app.js = 5 MB, load home.js, profile.js, dashboard.js

Benefits:

- Faster loading, better performance, smaller bundles

Q195. What is Tree Shaking?

Tree shaking removes unused code from final production bundles.

```
import { add } from "../math"
// Unused functions are removed during build.
```

Benefits:

- Smaller bundle size, faster application

Q196. What is SSR in Next.js?

SSR stands for Server-Side Rendering. In SSR, HTML is generated on the server before being sent to the browser.

Benefits:

- Better SEO, faster initial page load, improved performance

Flow:

Browser -> Server -> Rendered HTML -> Browser

Q197. What is Hydration in React?

Hydration is the process where React attaches event handlers to server-rendered HTML.

Flow:

Server Render HTML -> Browser Loads HTML -> React Adds Interactivity

Benefits:

- Interactive pages, better user experience

Q198. What is Webpack?

Webpack is a module bundler that combines project files into optimized bundles.

Responsibilities:

- Bundling files, minification, code splitting, asset optimization

```
module.exports = {
  entry: './src/index.js'
}
```

Q199. What is Babel?

Babel is a JavaScript compiler that converts modern JavaScript into browser-compatible JavaScript.

Benefits:

- Browser compatibility, support for modern JavaScript features

Q200. What is TypeScript and Why Use It?

TypeScript is a superset of JavaScript developed by Microsoft. It adds static typing, interfaces, and better tooling.

```
// JS
let age = 25

// TS
let age: number = 25
```

Benefits:

- Fewer bugs, better code quality, improved IntelliSense, easier maintenance